

## **Security Information and Event Management (SIEM) Log Analysis Using Splunk**

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### **Author Note**

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## Introductions

SIEM systems enable organizations to collect, centralize, monitor, and analyze security logs generated by computers, applications, and network devices. Utilizing tools such as *Splunk Enterprise* can make suspicious activity, incidents, and responding to threats be quickly identified by security analysts.

The purpose of this project was to build a barebones SIEM environment using Splunk Enterprise on macOS. With the use of sample logs that were imported to Splunk, I was able to simulate common system events. The services provided by Splunk were applied to create a (SOC) environment as a means of how events are analyzed.

## Project Objectives

The objectives of this project are as follows...

1. Install Splunk Enterprise
2. Import security log data
3. Search and analyse authentication events
4. Identify failed login attempts
5. Create statistical reports
6. Build a security monitoring dashboard
7. Configure a security alert

### Lab Environment

| Item             | Value                    |
|------------------|--------------------------|
| Operating System | macOS                    |
| SIEM Platform    | Splunk Enterprise        |
| Data Source      | Sample Security Log File |
| Index            | main                     |
| Project Type     | Security Log Analysis    |

### Methodologies

The following was performed...

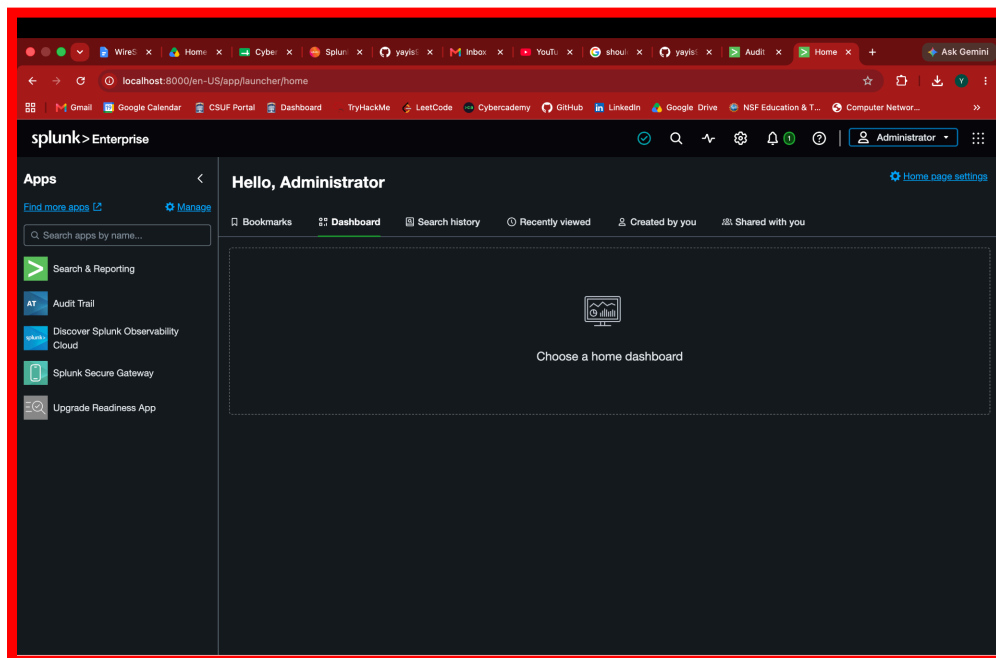
1. Installed Splunk Enterprise
2. Logged into the interface
3. Imported a sample security log file into the main index
4. Verified successful ingestion of log events
5. Executed multiple searches to identify authentication events
6. Filtered failed and successful login attempts
7. Generated statistical summaries
8. Created a security dashboard
9. Configured a security alert
10. Conducted a simulated incident investigation

### Findings and Analysis

#### 5.1 Splunk Installation

Splunk Enterprise was installed and configured appropriately on macOS. The web interface was accessed through the local Splunk server.

## Splunk Home Dashboard

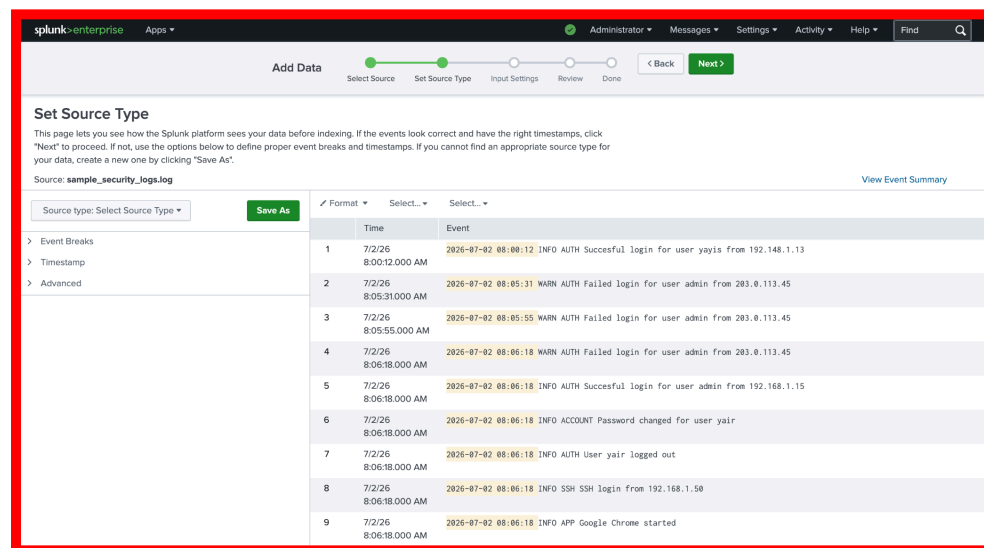


**Figure 1.** Splunk Enterprise dashboard used for centralized security monitoring and event analysis

## 5.2 Log Data Import

A sample Security log file containing authentication events, application activity, SSH events, etc.

When file insertion was complete Splunk indexed the events, making them searchable.



**Figure 2.** Importing security log data into Splunk for centralized event collection

## 5.3 Event Search

A search was performed using:

```
</> spl
```

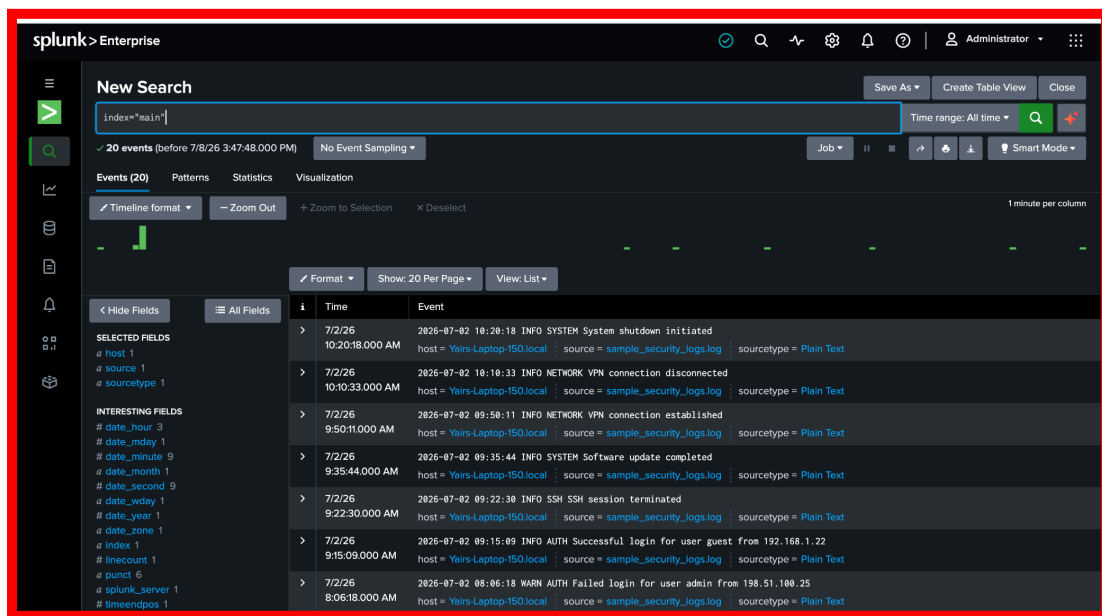
```
index=main
```

This search displayed all imported security events in chronological order. These events contained timestamps, event security levels, and descriptions of system activity.

The import data included:

- Successful Logins

- Failed Login
- User logout
- Password changes
- SSH login events
- Application startup and shutdown



## 5.4 Authentication Analysis

Authentication events were analyzed using the search:

```
</> spl
```

```
index=main "Successful login"
```

The results identified successful user authentication.

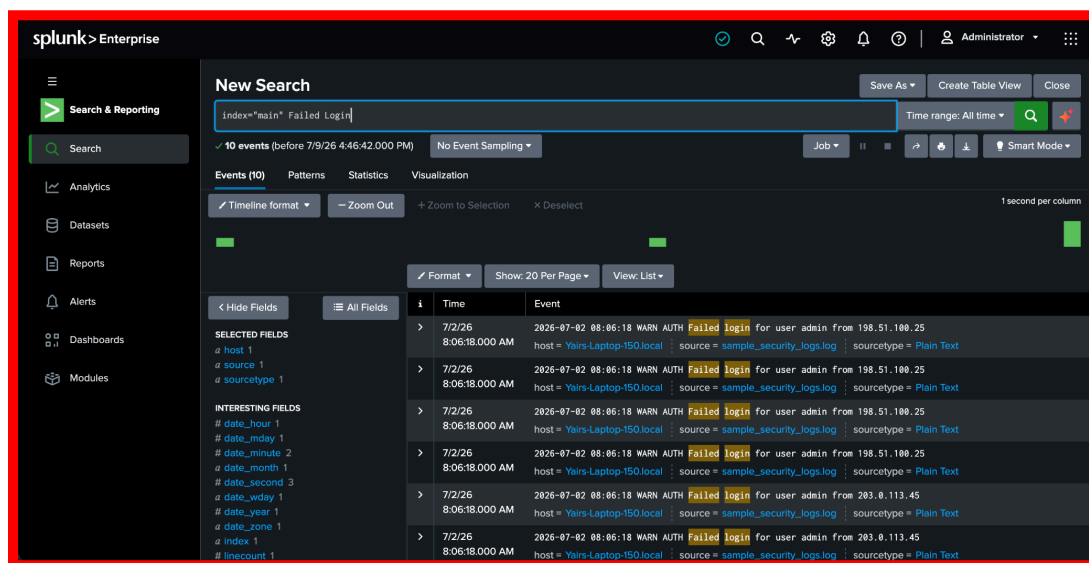
A second search was performed after the first initial search:

```
</> spl
```

```
index=main "Failed Login"
```

This search identified multiple failed authentication attempts involving different user accounts.

The repeated failed login attempts demonstrate how Splunk can identify suspicious authentication activity.



## 5.5 Statistical Analysis

Statistical analysis was performed using Splunk Search Processing Language

```
</> spl
```

```
index=main
```

```
| rex"^\S+\s+\S+\s+(?<level>\w+)"
```

```
| stats count by level
```

This query extracted the log severity level and generated a count of events by severity. The statistics provided an overview of information and warning events.

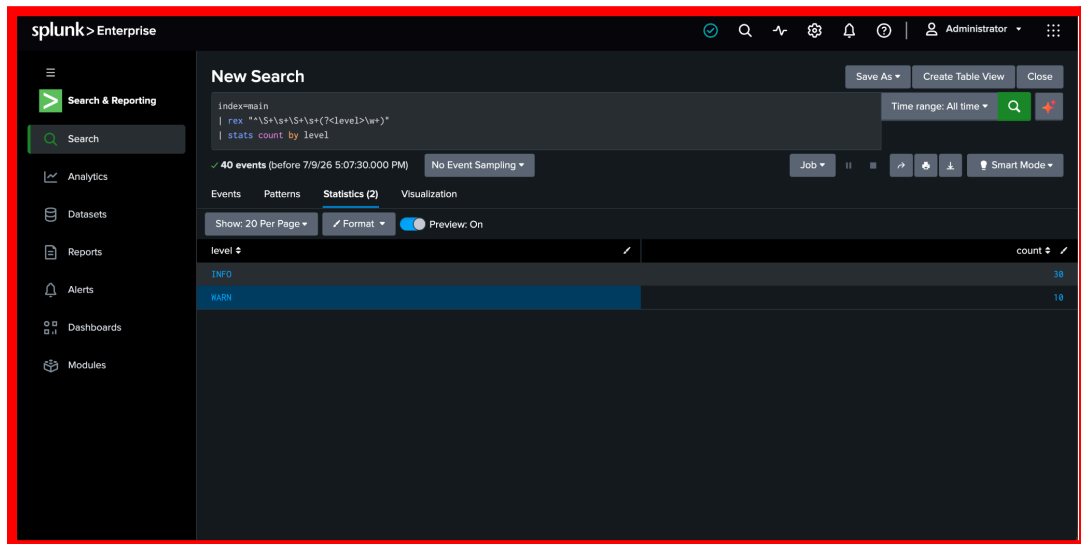


Figure 5. Statistical summary of security events generated using Splunk

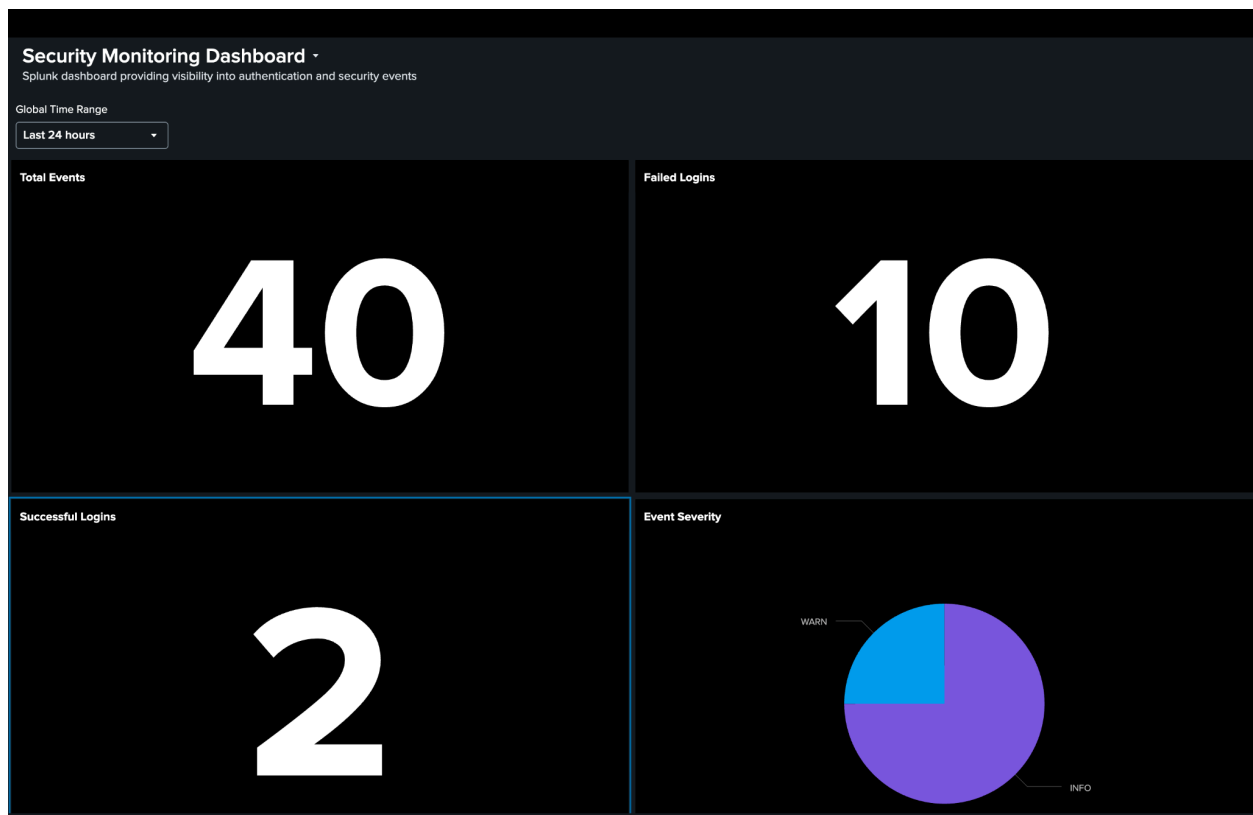


## 5.6 Dashboard Creation

A dashboard was created to provide visibility into authentication events and overall system activity.

Dashboard includes:

- Total Security Events
- Failed Login Attempts
- Successful Login Events
- Authentication Activity



**Figure 6.** Security dashboard providing centralized monitoring of authentication activity

## 5.7 Alert Configuration

A splunk alert was created to detect excessive failed login attempts

Search used:

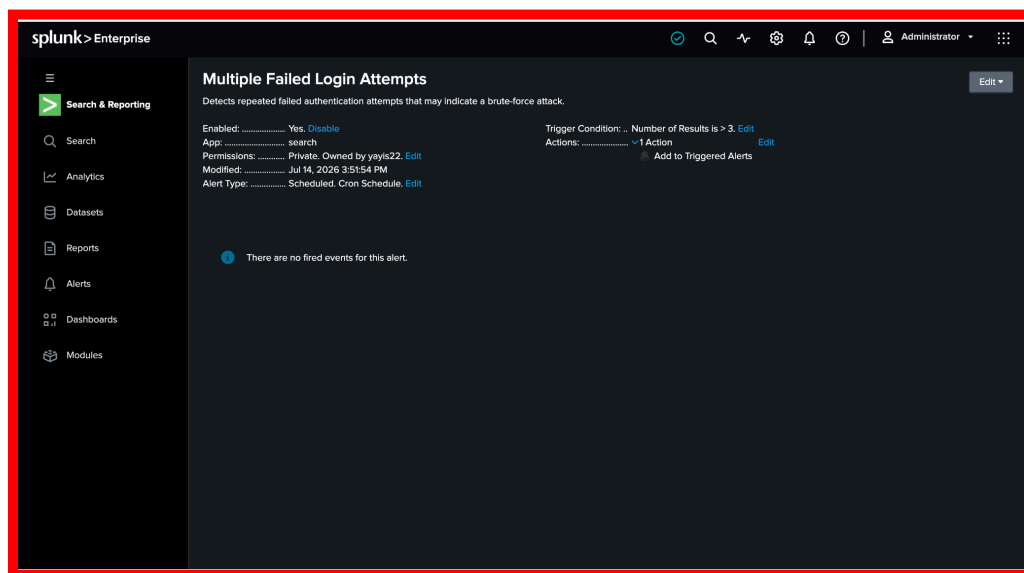
```
</> spl
```

```
index=main "Failed login"
```

Alert configuration:

- Trigger if more than three failed events occur within five minutes

This alert simulates how SOC analysts detect potential brute-force attacks or unauthorized login attempts



**Figure 7.** Splunk alert configured to detect repeated use of the imported log data.

## Security Findings

Upon review of the analysis here is what was found:

- Security events were successfully collected
- Authentication activity was easily searchable using SPL queries
- Multiple failed login attempts were identified
- Statistical summaries improved event visibility
- Dashboards simplified security monitoring
- Alerts provided automated detection of suspicious authentication behavior

### **Security Recommendations**

Based on the analysis, the following recommendations are provided:

- Enable MFA for all account
- Investigate repeated failed logins
- Configure automated alerts for abnormalities
- Regularly review SIEM dashboards for trends
- Maintain centralized logging across all systems
- Retain security logs according to organizational policies
- Update Splunk detection rules to identify additional attack techniques